

MSc Econometrics and Mathematical Economics (EME)

Kick-off session for cohort entering in Fall 2026

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Econometrics and Mathematical Economics

- ▶ Goal of the program: enable you to do meaningful work in your future job.
- ▶ Aim is to combine knowledge of econometrics with understanding of economics and/or business problems.
- ▶ We use statistics/mathematics as tools for understanding the real world and making better decisions.

Component	Cluster	ECTS
Core courses	A	24
Electives	B	18
Master Thesis	–	18
Total		60

1 course/elective = 6 ECTS

Cluster A courses (Core Courses): 24 ECTS

You must take 4 of the following:

Fall semester

- ▶ Microeconometrics (Units 1-2)
- ▶ Panel Data Analysis of Microeconomic Decisions (Units 1-2)
- ▶ Empirical Finance (Unit 1)
- ▶ Market Design (Unit 2)

Spring semester

- ▶ Data Science Methods (Unit 3)
- ▶ Dynamic Economic Models (Units 3-4)
- ▶ Time Series and their Applications (Unit 4)

Cluster B courses (Electives): 18 ECTS

- ▶ Many additional courses to choose from (and this list is not exhaustive):
 - ▶ Machine Learning Analytics
 - ▶ Individual Decision Making
 - ▶ Macroeconomic Dynamics: Consumption, Investment and Growth
 - ▶ Financial Markets and Institutions
 - ▶ Valuation Risk and Management
 - ▶ Decision Making under Uncertainty
 - ▶ Economics and Finance of Pensions
 - ▶ Asset Liability Management
 - ▶ Professional Business Analytics Skills
 - ▶ Games and Cooperative Behavior
 - ▶ Decentralized Finance and Blockchains
 - ▶ Dynamic Real Investment
- ▶ To help you make your choice, we present job profiles with corresponding core courses and electives.
- ▶ You can ask for additional courses in the study plan.

Master Thesis (18 ECTS)

- ▶ You can choose your own topic, and choose your own start date.
- ▶ You can write a thesis with a company in an internship, or you can write an internal thesis.
- ▶ Thesis internship topics:
 - ▶ Canvas page for thesis BAOR/QFAS/EME
 - ▶ Career Services
 - ▶ Asset Econometrics
 - ▶ Own networks.
- ▶ Internal thesis topics:
 - ▶ Canvas page for thesis BAOR/QFAS/EME
 - ▶ Professors
 - ▶ Own ideas.
- ▶ Topic must relate to EME core courses (i.e. Economics, Econometrics, Finance).

Master thesis (18 ECTS)

- ▶ Write an email with a brief proposal to the thesis coordinator.
- ▶ Thesis coordinator: Denis Kojevnikov.
- ▶ Thesis coordinator approves the topic and assigns a supervisor.
- ▶ Thesis has to be completed in 6 “thesis months”.
- ▶ You can start as soon as you have 30 ECTS.
- ▶ Start looking for a topic or internship early.

Microeconometrics (Fall)

- ▶ Wide range of methods for analysis of individual-level cross-sectional data:
 - ▶ Estimation of causal effects.
 - ▶ Structural estimation.
 - ▶ Semi- and nonparametric methods.
 - ▶ Models for binary, ordered, and multinomial choice.
- ▶ Many applications.
- ▶ Good preparation for working with “big data”.
- ▶ Complementary to the Panel Data Analysis course, but can be taken independently.

Panel Data Analysis of Microeconomic Decisions (Fall)

- ▶ Empirical models to analyse panel data on decisions of economic agents: individuals, households, and firms.
- ▶ Complement to Microeconometrics course, but can also be taken independently.
- ▶ Static and dynamic linear and non-linear models.
- ▶ Applications to consumption, savings and investment, labor economics, retirement and pensions, health economics, development, and environmental economics.

Empirical Finance (Fall)

- ▶ Empirical applications in finance:
 - ▶ The Efficient Market Hypothesis.
 - ▶ Predictability of asset returns.
 - ▶ Climate finance.
 - ▶ The finance of pensions and aging.
 - ▶ Behavioral finance.
- ▶ Econometric techniques: linear regression, generalized method of moments, maximum likelihood.

Market Design (Fall)

- ▶ Theory and application of market design without money (matching) and with money (auctions).
- ▶ Matching: examples include kidney exchange and school admissions.
- ▶ Auctions: examples include broadband spectrum auctions and internet advertising and keyword auctions.

Data Science Methods (Spring)

- ▶ Focus on modern statistical and machine learning techniques to make business decisions that involve big data.
- ▶ Applications in finance, marketing and economics.
- ▶ Cleaning, summarizing and visualizing big data.
- ▶ Model selection for prediction: LASSO, ridge, factor models, trees, random forests.
- ▶ Model selection for classification: clustering, trees, random forests, support vector machines.

Dynamic Economic Models (Spring)

- ▶ Emphasis on dynamic decision models under uncertainty.
- ▶ Markov decision problems and Markov games in discrete time, with or without strategic interaction between players.
- ▶ Examples from life-cycle analysis, demand estimation, learning, competition, and antitrust policies.
- ▶ Great combination of economic theory, econometric methods, and computation.

Time Series and their Applications (Spring)

- ▶ Estimation, testing, and forecasting with time series models.
- ▶ Examples: cash flows, profits, macro variables like unemployment, GDP, and interest rates.
- ▶ Focus on time series econometric models and techniques: ARMA, VAR, ARCH, SV, dynamic regression, Kalman filtering.
- ▶ Used by central banks, pension funds, insurers, and policy institutions like CPB.

Possible Elective Courses

You choose 3×6 ECTS courses among:

- ▶ Remaining core courses.
- ▶ Courses from MSc QFAS and MSc BAOR.
- ▶ Courses in Cooperative Game Theory and Dynamic Real Investment.
- ▶ Courses from Research Master Economics (subject to approval by CentER).
- ▶ Courses from MSc Economics (maximum one course).
- ▶ Data science courses in MSc Economics, MSc Finance, and MSc BAOR.

Software workshops and DataCamp

Software Workshop

- ▶ The program offers video lectures with introductions to Matlab and Stata.
- ▶ Canvas page for software courses.
- ▶ There can be Q&A sessions on the video lectures.

DataCamp

- ▶ This online learning environment features courses in R, Python and SQL.
- ▶ Every student can learn to program in a user-friendly way, free of charge, and without any prior knowledge.
- ▶ You can start at any moment and take courses at your own pace.

Electives and Job Profiles

- ▶ The MSc EME program can prepare you for a number of different job profiles.
- ▶ We present job profiles with corresponding core courses and electives for:
 - ▶ Economic Policy Analyst.
 - ▶ Quantitative Financial Analyst.
 - ▶ Data Scientist.

Note on Acronyms:

- ▶ RME: Research Master Economics (subject to approval by CentER).
- ▶ RMB: Research Master Business (subject to approval by CentER).
- ▶ MSE: MSc Economics (at most one elective).

Job Profile: Economic Policy Analyst

Typical first jobs:

Economist, analyst, policy advisor, or researcher with consultancy companies, banks, research organizations, the government, and the public sector.

Related programs:

Taking courses in this sequence helps facilitate a later switch to the Research Master Economics.

Job Profile: Economic Policy Analyst

Fall semester

- ▶ Market Design
- ▶ Panel Data Analysis of Microeconomic Decisions
- ▶ Microeconometrics
- ▶ Macroeconomics Dynamics: Consumption, Investment, and Growth (RME)
- ▶ Individual Decision Making (Microeconomics) (RME)
- ▶ Financial Markets and Institutions (MSE)
- ▶ Growth and Regional Development (MSE)
- ▶ Econ. and Psych. of Risk and Time (MSE)
- ▶ Health Economics (MSE)

Spring semester

- ▶ Time Series Models and their Applications
- ▶ Data Science Methods
- ▶ Dynamic Economic Models
- ▶ Games and Cooperative Behavior
- ▶ Dynamic Real Investment
- ▶ Core Courses in research master economics (RME)
- ▶ Environmental and Resource Economics (MSE)
- ▶ Labor Economics (MSE)
- ▶ Data Science for Economics (MSE)

Job Profile: Quantitative Financial Analyst

Typical first jobs:

Quantitative financial analyst, risk manager.

Related programs:

Stronger emphasis on econometrics than the QFAS Master.

Job Profile: Quantitative Financial Analyst

Fall semester

- ▶ Empirical Finance (Unit 1)
- ▶ Microeconometrics
- ▶ Panel Data Analysis of Microeconomic Decisions
- ▶ Valuation and Risk Management
- ▶ Financial Markets and Institutions (MSE)

Spring semester

- ▶ Time Series and their Applications (Unit 3)
- ▶ Data Science Methods (Unit 3)
- ▶ Asset Liability Management (Unit 3)
- ▶ Issues in Finance and Insurance
- ▶ Risk and Regulation
- ▶ Dynamic Real Investment (Unit 4)
- ▶ Professional Business Analytics Skills

Job Profile: Data Scientist

Typical first jobs:

Analyst or researcher in economics and business at management, service and consultancy companies, non-profit organizations, and application providers.

Related programs:

More focused on economics and econometrics than the BAOR Master.

Job Profile: Data Scientist

Fall semester

- ▶ Microeconometrics
- ▶ Panel Data Analysis of Microeconomic Decisions
- ▶ Empirical Finance (Unit 1)
- ▶ Machine Learning Analytics
- ▶ Decision Making under Uncertainty

Spring semester

- ▶ Data Science Methods (Unit 3)
- ▶ Time Series and their Applications
- ▶ Professional Business Analytics Skills
- ▶ Data Science for Economics (MSE)

Further topics

- ▶ Education Committee: mail the program coordinator if you would like to join.
- ▶ Sounding board: mail the program coordinator if you would like to join.
- ▶ Please fill out teaching evaluations.
- ▶ Come to lectures and tutorials.
- ▶ Reach out to other students.

After graduation

- ▶ Think early about the next steps.
- ▶ Attend career events.
- ▶ See list with recommended career events on Canvas.
- ▶ National Econometricians' Day: sign up early.
- ▶ Consider joining Asset Econometrics.
- ▶ Check Career Services.

Education Coordinator



First point of contact for:

- ▶ Study progress & study related issues
- ▶ Study advice & planning
- ▶ Special personal circumstances
- ▶ Extracurricular activities

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Also feel free to get in touch with me!

- ▶ Feedback on the program
- ▶ Advice on next steps after the EME
 - ▶ Especially if interested in an academic career